

Obstacle Challenge Navigation Algorithm

Overview

The Obstacle Challenge extends the Open Challenge by introducing colored obstacles that must be avoided while maintaining autonomous navigation.

The robot combines:

- Three HC-SR04 ultrasonic sensors
- MPU6050 gyroscope
- ESP32-CAM vision system
- MG90S steering servo
- Two DC drive motors
- Arduino Uno

The objective is to complete the required laps while correctly passing red and green obstacles.

Sensor Configuration

Front Ultrasonic Sensor

Used to:

- Detect obstacles ahead
- Measure safe passing distance
- Trigger avoidance maneuvers

Left Ultrasonic Sensor

Used to:

- Monitor distance from left wall
- Assist during obstacle avoidance

Right Ultrasonic Sensor

Used to:

- Monitor distance from right wall
- Assist during obstacle avoidance

MPU6050

Used to:

- Measure heading changes
- Verify turns
- Improve maneuver accuracy

ESP32-CAM

Used to:

- Detect obstacle colors
- Identify red obstacles
- Identify green obstacles
- Communicate obstacle information to Arduino

Obstacle Detection Logic

Green Obstacle

When a green obstacle is detected:

Pass on the RIGHT side

The robot adjusts its path to move around the obstacle while maintaining track awareness.

Red Obstacle

When a red obstacle is detected:

Pass on the LEFT side

The robot modifies its trajectory to safely avoid the obstacle.

Navigation Strategy

Normal Driving

The robot follows the wall using ultrasonic sensors exactly as in the Open Challenge.

```
Follow wall
↓
Detect corner
↓
Turn
↓
Continue
```

Obstacle Encounter

When the front ultrasonic detects an object within a predefined distance:

```
Obstacle detected
```

The robot:

1. Reduces speed.
2. Requests obstacle color from ESP32-CAM.
3. Determines avoidance side.
4. Executes avoidance maneuver.
5. Returns to normal navigation.

Green Obstacle Maneuver

```
Obstacle detected
↓
Color = Green
↓
Move Right
↓
Pass obstacle
↓
Return to center path
```

Red Obstacle Maneuver

```
Obstacle detected
↓
```

```
Color = Red
↓
Move Left
↓
Pass obstacle
↓
Return to center path
```

Corner Priority

Track navigation has higher priority than obstacle avoidance.

If a corner and obstacle are detected simultaneously:

1. Complete the corner safely.
2. Re-evaluate obstacle position.
3. Execute avoidance maneuver.

Gyroscope Usage

The MPU6050 is used to:

- Maintain heading during obstacle avoidance.
- Measure steering corrections.
- Return accurately to the original trajectory.

Pseudocode

```
Initialize Sensors

while(race_not_finished)

    Read Ultrasonic Sensors

    Follow Wall

    if ObstacleDetected()

        Color = ReadCamera()

        if Color == RED
```

```
AvoidLeft()

else if Color == GREEN

    AvoidRight()

    if CornerDetected()

        Turn90Degrees()

Stop Motors
```

ESP32-CAM Communication

ESP32-CAM sends simple commands to Arduino:

```
R = Red Obstacle

G = Green Obstacle

N = No Obstacle
```

Example:

```
R
```

Arduino executes:

```
AvoidLeft()
```

Example:

```
G
```

Arduino executes:

```
AvoidRight()
```